

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

22SAM21

Second Semester M.Tech. Degree Examination, June/July 2024 Data and Web Mining

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Outline the unique characteristics of the Web.	8	L2	CO1
	b.	Explain the steps of Prefix span algorithm for multiple minimum supports.	12	L1	CO1

OR					
Q.2	a.	Describe the main steps in Data mining.	10	L2	CO1
	b.	Explain the different tasks of Web mining in detail.	10	L2	CO1

Module – 2					
Q.3	a.	Illustrate the basic learning process of the classification model with the help of a suitable figure.	10	L3	CO2
	b.	Explain in brief the use of decision tree for classification and construct the decision tree for the following loan application data set given in the below table : 1.	10	L2	CO2

ID	Age	Has Job	Own House	Credit rating	Class
1	Young	False	False	Fair	No
2	Young	False	False	Good	No
3	Young	True	False	Good	Yes
4	Young	True	True	Fair	Yes
5	Young	False	False	Fair	No
6	Middle	False	False	Fair	No
7	Middle	False	False	Good	No
8	Middle	True	True	Good	Yes
9	Middle	False	True	Excellent	Yes
10	Middle	False	True	Excellent	Yes
11	Old	False	True	Excellent	Yes
12	Old	False	True	Good	Yes
13	Old	True	False	Good	Yes
14	Old	True	False	Excellent	Yes
15	Old	False	False	Fair	No

Table : 1. A loan application data set.

OR					
Q.4	a.	Illustrate the GSP algorithm for generating sequential patterns.	8	L3	CO2
	b.	Describe in brief about the sequential rules with a suitable example.	12	L2	CO2

Module – 3					
Q.5	a.	Explain the three inverted index method steps applied in documents searching.	10	L2	CO3
	b.	Describe about Single – link and Average – link method with the help of a neat diagram.	10	L2	CO3
OR					
Q.6	a.	Describe in brief about the user query and document retrieval process of singular value decomposition.	10	L2	CO3
	b.	Discuss in brief on Searching and Ranking on a user query.	10	L3	CO3
Module – 4					
Q.7	a.	Illustrate the concept of Co-citation and Bibliographic coupling in link analysis.	10	L3	CO4
	b.	Describe the Breadth First Crawlers in detail.	10	L2	CO4
OR					
Q.8	a.	Explain the Hyperlink Induced Topic Search (HITS) algorithm to discover and rank the webpages.	10	L2	CO4
	b.	Describe the types of Web Crawlers in detail.	10	L2	CO4
Module – 5					
Q.9	a.	Explain in brief on the classification based sentiment phrases.	12	L2	CO5
	b.	Illustrate the data collection and preprocessing in Web Usage Mining method.	8	L3	CO5
OR					
Q.10	a.	Explain the various Text classification methods in Web mining.	10	L2	CO5
	b.	Describe about the Cluster Analysis and Visitor Segmentation in Web usage mining.	10	L2	CO5